

UNIT 2

Mathematical Reasoning

About the Test

The GED® *Mathematical Reasoning Test* evaluates your ability to solve mathematical problems and apply mathematics to a variety of contexts. You will have 90 minutes to answer approximately 46 questions based on the following:

- **Quantitative Reasoning (45%)**, which includes problems with positive and negative whole numbers, decimals, and fractions; ratios, proportions, and percents; data and statistics; and geometric measurement
- **Algebraic Reasoning (55%)**, which includes expressions, polynomials, equations, inequalities, linear equations, quadratic equations, and patterns and functions. A variety of formats will assess evaluating, solving, and graphing skills in algebra.

You will need a minimum score of 145 to pass the *Mathematical Reasoning Test*, which is one of the four tests you will need to pass in order to earn a high school equivalency diploma.

Tools

Calculator For most of the test, you will be able to use either a handheld or an online version of the **Texas Instruments TI-30XS MultiView™ scientific calculator**. You can access the online calculator by clicking on this icon. You can also take your own handheld TI-30XS MultiView™ calculator to use on test day.



For the first five questions on the test, you will not be able to use the calculator. You can take up to 12 of the 90 minutes with these questions. Check all of your work carefully before you leave this section of the test. You will not be able to return to these problems once the online calculator is enabled.

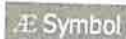
If you forget how to use a specific calculator function, you will be able to access a **Calculator Reference Sheet** from the computer screen by clicking on this icon.



Formula Sheet A copy of this formula sheet appears on page 413 of this book. You will be able to access these formulas by clicking on this icon. Other formulas that appear in this book should be memorized.



Symbol Selector On some fill-in-the-blank problems, you will need to use a symbol from the symbol selector toolbar. You can enable it when the icon to the right shows up on the light blue toolbar at the top of your screen.



At the testing center, you will be given a dry erase marker and a **wipe-off board** that you can use as scratch paper.

Computer-Based Mathematics Test Formats

Many of the questions will be *multiple-choice* with four options (A through D). You will **click** on the best answer.

A painter needs to cover the walls of four rooms in two coats of paint. Each room measures 4 ft by 8 ft, and the ceiling is 10 feet high in each. Each gallon of paint covers 32 square feet. How many gallons of paint does he need?

- ☐ A. 2
- ☐ B. 60
- ☐ C. 128
- ☐ D. 256

GED® TEST TIP

Follow all directions carefully. For example, in this *fill-in-the-blank* problem, you are asked to **type** the answer, **simplify the expression**, and **use the symbol selector** to complete the result.

Fill-in-the-blank questions will require you to **type** a numerical answer to a problem or to enter an equation by **clicking** to open the symbol selector and then **inserting** the symbol in the box. You may also be asked to **type** a word or phrase in the box to answer a question about your thought process.

Type your answer in the box. You may use numbers, symbols, and/or text in your answer.

Simplify the expression completely. Leave your answer in radical form.

$$\sqrt{60} \div \sqrt{5}$$

(NOTE: Click the symbol selector when you need to insert the radical sign.)

Drop-down questions give you the opportunity to **click** the word "Select" and choose the option that correctly completes the statement.

Sarah runs an open babysitting service on Saturday nights so that parents in the neighborhood can enjoy a night out. The chart below shows the number of children in attendance each hour.

Hour	# of children
5 p.m.	2
6 p.m.	4
7 p.m.	5
8 p.m.	6
9 p.m.	5
10 p.m.	2
11 p.m.	2

The number of children in attendance ranged from Select...
to Select...

Drag-and-drop items require you to **click** on an image, number, or symbol and to **drag** and **drop** it to a specified location on the computer screen to create an expression, equation, or inequality.

A clown creates an equation that relates his data of each child's head size to the length of the balloon he needs to make them a proper balloon-animal hat. For each child, he needs a balloon five times as long as the child's head size, plus $\frac{4}{5}$ of an inch extra. Complete a linear equation for child A, where x is the size of the child's head in inches and y is the length of the balloon in inches.

Click on the variables and numbers you want to select and drag them into the boxes.

Equation for Child A

$$y = \square \square + \square$$

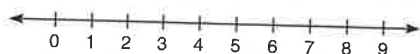
0.7	0.8	-0.7	-0.8
5	5.2	5.4	x

Hot-spot questions are based on a graphic such as a number line, coordinate grid, or geometric figure. You will need to **click** on a specific place or places on the figure.

Jessica has 20 hours a week to spend making scarves. Each week, she spends 2 hours purchasing the materials. Each scarf takes 3 hours to complete. The inequality $2 + 3s \leq 20$ represents the number of scarves, s , Jessica can make in one week.

Using the number line below, click each number to plot each possible value for s .

(NOTE: If you need to remove a point, hold the cursor over the point and left-click the mouse.)



GED® TEST TIP

You may need to click on one point or on more than one point to correctly answer a hot-spot question. Check the finished graphic against the question before moving on. Make sure that you have completed the task by creating all required points.

GED® TEST TIP

You are encouraged to purchase a handheld version of the Texas Instruments TI-30XS MultiView™ calculator for practice as you work through this book. You can buy one at office or school supply stores or online.

Using the Texas Instruments TI-30XS MultiView™ Calculator

Throughout this unit, you will practice using the TI-30XS MultiView™ calculator. Remember: You do not have to use the calculator to solve problems; only use it if it is relevant to the problem or if you believe it will be helpful.

Below is a picture of the handheld version and some issues to keep in mind.

- The handheld version has an On/Off button at the lower-left hand side.
- To use the online version of the calculator, click the calculator icon (see page 213) to open the calculator in a new window. Use your mouse to **drag** the calculator to the side of the computer screen so that it does not cover the problem. When you are finished with a calculation, click on the "X" at the top of the window to close the calculator.
- When you are finished with a calculation, use the **enter** button. (There is no "equal sign" button on this calculator.)
- Use the **white numeric keypad** to enter numbers, the decimal point, and the negative sign (–) for a negative number such as –13.
- The **operation keys**, on the right, allow you to add, subtract, multiply, and divide. (Use the minus sign here for subtraction; use the negative sign (–) on the keypad to indicate a negative number.)
- Use the bright green **2nd button** at the left of the calculator to activate the **2nd functions** (in bright green) above the keys. Some of the main ones are square root, exponent, percent, and mixed number functions.
- The **four arrows** at the top right of the calculator allow you to move the **onscreen cursor** up, down, left, or right as needed. The **delete** key to the left of the arrows allows you to correct mistakes as you work.
- Every time you begin a calculation, press the **clear** button above the **operation keys** to clear the calculator's memory.

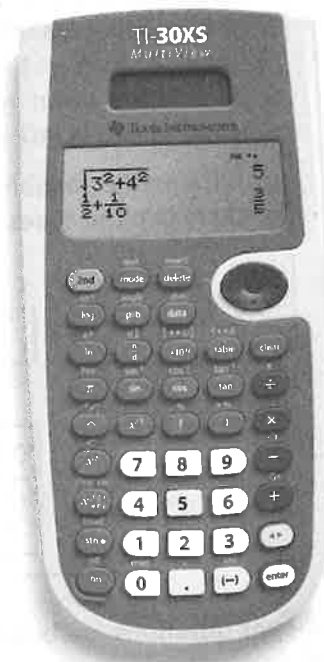


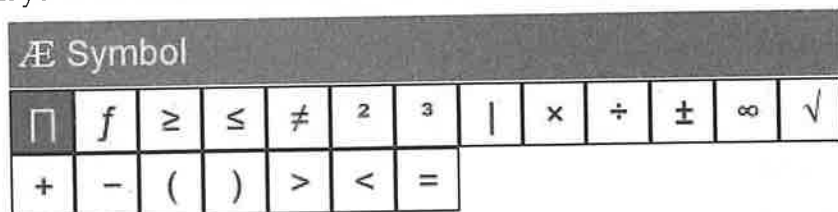
Photo used with permission from TI. <http://education.ti.com>.

TI-30XS MultiView™ Calculator

Accessing the Symbol Selector Toolbar

Some questions will instruct you to use a particular symbol from the “symbol selector.” This is not available for all problems, but this icon shows up when you need the toolbar: 

When you **click** on this icon, a new window will open up with these symbols:



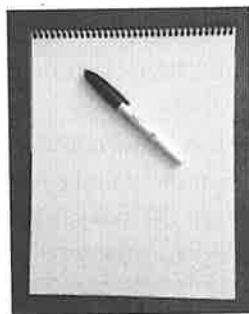
Click on a symbol to insert it in the correct place in an expression, equation, or inequality. When you are finished with the toolbar, **click** on the “X” at the top of the window to close the window.

Using the Wipe-Off Board

Even with all of the computer-based tools, you may still feel the need for some scratch paper to do the following:

- Write down the relevant numbers that you need to answer a specific question.
- Solve the problem by hand and use the calculator to check your work.
- Make a drawing or a diagram to help you picture a situation that is described in a word problem. This is an important strategy when working with measurement and geometry problems that do not contain a graphic.

At the testing center, you will be provided with a wipe-off board and dry erase marker, similar to the material shown at the right. Make use of the board throughout the *Mathematical Reasoning Test* and ask for a new one if you feel that you need it.



GED® TEST TIP

If you do not know all of these symbols now, copy them on a piece of paper or in a math notebook. Write meanings and examples for them as you encounter them in this book so you have mastery of them before the test.